

Basic Network Sniffer

CodeAlpha Cyber Security Internship — Task 1

1. Introduction

A network packet sniffer is a cybersecurity tool used to capture and analyze network traffic. It helps in understanding how data moves between devices and how different network protocols communicate.

In this project, a basic network sniffer was developed using Python and the Scapy library.

2. Objective

The main objectives of this project are:

- To capture network packets in real time.
- To analyze packet information.
- To identify different network protocols.
- To extract useful details from network traffic.

3. Tools Used

- Python 3
- Scapy
- Kali Linux

4. Implementation

The program uses Scapy's `sniff()` function for continuous packet capturing.

For every captured packet, the program checks:

- Source IP address
- Destination IP address
- Protocol type
- Packet length
- Payload information

The program identifies TCP, UDP, and ICMP packets and displays their details.

5. Features Implemented

Source IP Detection

The program extracts the IP address of the device sending the packet.

Destination IP Detection

The program displays the destination address where the packet is being sent.

Protocol Identification

The sniffer identifies:

- TCP
- UDP
- ICMP

Packet Length

The size of every captured packet is displayed in bytes.

Payload Extraction

If packet data is available, the payload information is displayed.

Continuous Monitoring

The program keeps capturing packets until manually stopped by the user.

6. Testing

The program was tested by running:

```
sudo python3 sniffer.py
```

Network traffic was generated using:

```
ping google.com
```

The sniffer successfully displayed packet details.

7. Result

The developed network sniffer successfully captured and analyzed live network packets. It displayed important packet information including IP addresses, protocols, packet size, and payload data.

8. Conclusion

This project provided practical understanding of network traffic monitoring and packet analysis. It demonstrates the basic working principles of cybersecurity tools used for network analysis and troubleshooting.

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